# ME 5309 – Mid-Semester Examination

Answer Booklet

**Instructions:** Attempt any three questions. Each question carries 10 marks. Show working where appropriate.

## Question 1 [10 marks]

### Q1(a) [4 marks]

### Q1(b) [3 marks]

### Q1(c) [3 marks]

## Question 2 [10 marks]

### Q2(a) [2 marks]

The state model is $\dot{x}_1 = x_2$, $\dot{x}_2 = (\tau - 0.05x_2 - 2.4525 \sin x_1)/0.02$. Equilibrium requires $\tau_0 = 2.4525 \sin \theta_e$, so $\tau_0$ must be between -2.4525 and +2.4525 N m.

### Q2(b) [2 marks]

$$A = [[0, 1], [-61.3125, -2.5]], B = [[0], [50]]$$

### Q2(c) [4 marks]

The balance is $50(90 - \theta ss) = 2.4525 \sin \theta ss$. Solving gives a small error of about 0.05°.

### Q2(d) [2 marks]

The added gravity torque should cancel the load, so the steady-state error becomes zero.

## Question 3 [10 marks]

### Q3(a) [2 marks]

The estimate is noisy and can fluctuate from sample to sample.

### Q3(b) [2 marks]

$$y[k] = 0.8818y[k-1] + 0.0591x_k + 0.0591x[k-1], DCgain1$$

**Q3(c)** **[4 marks]**

**Q3(d)** **[2 marks]**

The observer is simpler because it only smooths the measured velocity. The filter is safer when the model is poor.

## Question 4 [10 marks]

**Q4(a)** **[2 marks]**

The closed-loop poles are -2±2j. I use Nff=c/b=6 for unity DC gain.

**Q4(b)** **[2 marks]**

$$\theta_{ss} = 0.75 \times 8/(6+2) = 0.75,$$

the percentage tracking error is 25%

**Q4(c)** **[4 marks]**

**Q4(d)** **[2 marks]**

Integral action gives zero steady-state error if stable. A practical problem is integrator windup.